

TrES-5b

R-2508

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_260720 · created 2026-08-24T23:59:17+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461241.7947 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.118 +/- 0.053
Transit depth (Rp/Rs)^2	1.39 +/- 1.26 [%]
Orbital Inclination (inc)	81.92 +/- 2.95
Airmass coefficient 1 (a1)	0.95 +/- 0.56
Airmass coefficient 2 (a2)	0.2 +/- 1.65
Scatter in the residuals of the lightcurve fit is	57.52 %
Best Comparison Star	#2 - [391, 242]
Optimal Aperture	1.03
Optimal Annulus	5.49
Transit Duration (day)	0.046 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.118 $\pm$ 0.053 vs 0.142 (deviation 0.45 σ)
Duration fit vs published	0.046 d vs 0.0758 d (deviation 1.03 σ)
Mid-time O-C	-10.4 min
Depth SNR	2.2
Residual scatter	57.52 %

Light curve

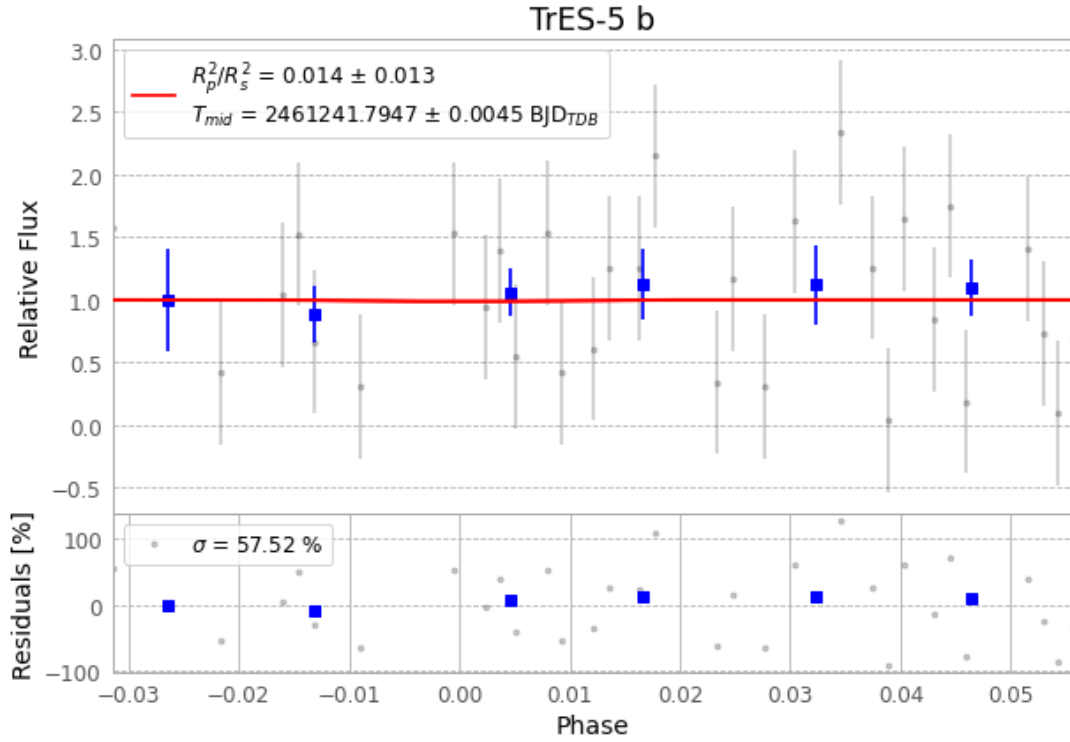


Figure 1 — FinalLightCurve_TrES-5 b_2026-07-19.png (SHA-256 4e01164d9377cd9b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [619, 193], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3dcd1f23bdfa3730...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), TRES-5260720055417.FITS → TRES-5260720090015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-260720.log (7e0f67055631...), FinalLightCurve_TrES-5 b_2026-07-19.png (4e01164d9377...), FinalLightCurve_TrES-5 b_2026-07-19.csv (f9f146a7058d...)

Compute resources

Wall time 113.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 23:59Z.